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SEMICONDUCTOR DEVICE AND METHOD OF MANUFACTURING THE SAME

Abstract

In one embodiment, a method of manufacturing a semiconductor device includes forming a first insulator. The method further includes forming a first layer including tungsten and nitrogen, on the first insulator. The method further includes reducing nitrogen concentration in the first layer by annealing the first layer in a state that the first layer is exposed. The method further includes forming a second layer including tungsten, on the first layer after the annealing.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2024-035254, filed on Mar. 7, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate to a semiconductor device and a method of manufacturing the same.

BACKGROUND

[0003] Regarding a semiconductor device including an interconnect layer such as an electrode layer, it is desired to reduce the electric resistance of the interconnect layer and suppress diffusion of atoms from the interconnect layer to another layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 is a perspective view showing a structure of a semiconductor device of a first embodiment;

[0005] FIGS. 2 to 5 are sectional views showing a method of manufacturing the semiconductor device of the first embodiment;

[0006] FIGS. 6A and 6B are sectional views showing a structure of a semiconductor device of a comparative example of the first embodiment;

[0007] FIG. 7 is a sectional view showing the structure of the semiconductor device of the first embodiment;

[0008] FIGS. 8A to 8C are sectional views showing details of the method of manufacturing the semiconductor device of the first embodiment;

[0009] FIGS. 9A and 9B are graphs concerning characteristics of the semiconductor device of the first embodiment;

[0010] FIG. 10 is another graph concerning characteristics of the semiconductor device of the first embodiment;

[0011] FIG. 11 is a sectional view showing a structure of a semiconductor device of a variation of the first embodiment;

[0012] FIGS. 12A and 12B are sectional views showing a structure of a semiconductor device of a second embodiment, and a structure of a semiconductor device of a variation of the second embodiment;

[0013] FIGS. 13A to 13D are sectional views showing a method of manufacturing the semiconductor device of the second embodiment;

[0014] FIG. 14 is a sectional view showing a structure of a semiconductor device of a third embodiment;

[0015] FIG. 15 is a sectional view showing a structure of a columnar portion of the third embodiment; and

[0016] FIGS. 16 and 17 are sectional views showing a method of manufacturing the semiconductor device of the third embodiment.

DETAILED DESCRIPTION

[0017] Embodiments will now be explained with reference to the accompanying drawings. In FIGS. 1 to 17, the same components are denoted by the same reference signs, and redundant description thereof will be omitted.

[0018] In one embodiment, a method of manufacturing a semiconductor device includes forming a first insulator. The method further includes forming a first layer including tungsten and nitrogen, on the first insulator. The method further includes reducing nitrogen concentration in the first layer by annealing the first layer in a state that the first layer is exposed. The method further includes forming a second layer including tungsten, on the first layer after the annealing.

First Embodiment

[0019] FIG. 1 is a perspective view showing a structure of a semiconductor device of a first embodiment. The semiconductor device of the present embodiment includes, for example, a three-dimensional semiconductor memory.

[0020] The semiconductor device of the present embodiment includes a core insulator 1, a channel semiconductor layer 2, a tunnel insulator 3, a charge storage layer 4, a block insulator 5, and an electrode layer 6. The block insulator 5 includes an insulator 5a and an insulator 5b. The electrode layer 6 includes a barrier metal layer 6a and an electrode material layer 6b. The electrode layer 6 is an example of the first interconnect layer. The barrier metal layer 6a is an example of a first layer. The electrode material layer 6b is an example of second and third layers and an example of an interconnect material layer in the first interconnect layer.

[0021] In FIG. 1, a plurality of electrode layers and a plurality of insulators are stacked alternately on a substrate, and a memory hole Ha is provided in the electrode layers and the insulators. FIG. 1 shows one electrode layer 6 of the electrode layers. The electrode layers function, for example, as word lines for a three-dimensional semiconductor memory. FIG. 1 shows an X direction and a Y direction parallel to a surface of the substrate and perpendicular to each other, and a Z direction perpendicular to the surface of the substrate. In the present specification, a +Z direction is treated as an upward direction and a -Z direction is treated as a downward direction. The -Z direction may or may not coincide with the gravity direction.

[0022] The core insulator 1, the channel semiconductor layer 2, the tunnel insulator 3, the charge storage layer 4, and the insulator 5a are formed in the memory hole Ha, making up a memory cell of the three-dimensional semiconductor memory. The insulator 5a is formed on side faces of the electrode layers and insulators in the memory hole Ha, and the charge storage layer 4 is formed on a side face of the insulator 5a. The charge storage layer 4 can store signal charges of the three-dimensional semiconductor memory. The tunnel insulator 3 is formed on a side face of the charge storage layer 4, and the channel semiconductor layer 2 is formed on a side face of the tunnel insulator 3. The channel semiconductor layer 2 functions as a channel of the three-dimensional semiconductor memory. The core insulator 1 is formed on a side face of the channel semiconductor layer 2. The insulator 5a is an example of a third insulator. The tunnel insulator 3 is an example of a fourth insulator.

[0023] The insulator 5a is, for example, a SiO.sub.2 film (silicon oxide film). The charge storage layer 4 is, for example, a SiN film (silicon nitride film). The tunnel insulator 3 is, for example, a SiO.sub.2 film. The channel semiconductor layer 2 is, for example, a polysilicon layer. The core insulator 1 is, for example, a SiO.sub.2 film.

[0024] The insulator 5b, the barrier metal layer 6a, and the electrode material layer 6b are formed between two of the plurality of insulators, i.e., formed on a lower face of the upper insulator, an upper face of the lower insulator, and the side face of the insulator 5a in sequence. The plurality of insulators and the block insulator 5 are examples of the first insulator. The plurality of insulators are also examples of a second insulator.

[0025] The insulator 5b is, for example, Al.sub.2O.sub.3 film (aluminum oxide film). The barrier metal layer 6a is, for example, a W (tungsten) layer including N (nitrogen) atoms as impurity atoms. The electrode material layer 6b is, for example, a W (tungsten) layer. Further details on the barrier metal layer 6a and the electrode material layer 6b will be described later.

[0026] FIGS. 2 to 5 are sectional views showing a method of manufacturing the semiconductor device of the first embodiment.

[0027] First, a substrate **11** is prepared, and a stacked film **12** including a plurality of sacrificial layers **13** and a plurality of insulators **14** alternately is formed on the substrate **11** (FIG. 2). The stacked film **12** is formed by stacking the plurality of sacrificial layers **13** and the plurality of insulators **14** alternately on the substrate **11**. The stacked film **12** may be formed either directly on the substrate **11** or formed on the substrate **11** via another layer. The substrate **11** is, for example, a semiconductor substrate such as a Si (silicon) substrate. The sacrificial layers **13** are, for example, SiN films. The insulators **14** are, for example, SiO₂ films. The sacrificial layers **13** are an example of a fourth layer. The insulators **14** are examples of first and second insulators.

[0028] Next, a plurality of memory holes Ha are formed in the stacked film **12** by photolithography and RIE (reactive ion etching) (FIG. 2). FIG. 2 shows one of the memory holes Ha. Each of the memory holes Ha of the present embodiment has a circular shape in planar view and penetrates the stacked film **12**.

[0029] Next, the insulator **5a**, the charge storage layer **4**, the tunnel insulator **3**, the channel semiconductor layer **2**, and the core insulator **1**, are formed in sequence on a side face of the stacked film **12** in each of the memory holes Ha (FIG. 3). The insulator **5a**, the charge storage layer **4**, the tunnel insulator **3**, and the channel semiconductor layer **2** are formed to have a tubular shape extending in the Z direction. The core insulator **1** is formed to have a columnar shape extending in the Z direction.

[0030] Next, a plurality of slits are formed in the stacked film **12** (not **10** shown) and the sacrificial layers **13** are removed from the slits using a chemical solution such as a phosphoric acid aqueous solution. As a result, a plurality of concave portions Hb are formed in the stacked film **12** (FIG. 4). The concave portions Hb are an example of first concave portions.

[0031] Next, the insulator **5b**, the barrier metal layer **6a**, and the electrode material layer **6b** are formed in sequence on surfaces of the insulators **5a** and **14** in each of the concave portions Hb (FIG. 5). As a result, the block insulator **5** including the insulators **5a** and **5b** is formed.

Furthermore, the electrode layer **6** including the barrier metal layer **6a** and the electrode material layer **6b** is formed in each of the concave portions Hb. Furthermore, the stacked film **12** including a plurality of the electrode layers **6** and a plurality of insulators **14** alternately is formed on the substrate **11**. In this way, a replacement process is performed to replace the sacrificial layers **13** with the electrode layers **6**. The insulators **14** and the block insulators **5** are examples of the first insulator.

[0032] Each of the concave portions Hb is formed between a pair of the insulators **14** adjacent to each other in the Z direction. In each of the concave portions Hb, the insulator **5b**, the barrier metal layer **6a**, and the electrode material layer **6b** are formed on the lower face of the upper insulator **14**, the upper face of the lower insulator **14**, and the side face of the insulator **5a** in sequence. As a result, each of the electrode layers **6** is formed between the insulators **14** via the insulator **5b**. On the other hand, after the replacement process, in each of the memory holes Ha, the charge storage layer **4** is formed on a side face of the plurality of electrode layers **6** via the insulator **5a** and the channel semiconductor layer **2** is provided on the side face of the charge storage layer **4** via the tunnel insulator **3**.

[0033] In this way, the semiconductor device of the present embodiment is manufactured (FIG. 5). FIG. 1 shows part of the semiconductor device shown in FIG. 5.

[0034] Next, the first embodiment will be compared with a comparative example with reference to FIGS. 6A, 6B, and 7.

[0035] FIGS. 6A and 6B are sectional views showing a structure of a semiconductor device of a comparative example of the first embodiment.

[0036] FIG. 6A shows the structure of the semiconductor device of the present comparative example. As with FIG. 5, FIG. 6A shows the charge storage layer **4**, the block insulator **5**, the electrode layer **6**, the insulators **14**, and the like. However, the electrode layer **6** of the present comparative example includes a barrier metal layer **6a'** instead of the barrier metal layer **6a**. FIG.

6A further shows a seam Ga formed in the electrode material layer 6b during formation of the electrode material layer 6b. The seam Ga is an air gap extending generally in parallel to an X-Y plane. As shown in FIG. 6A, the electrode material layer 6b of the present comparative example includes metal layers 21 and 22 formed on a side face, upper face, and lower face of the barrier metal layer 6a' in sequence.

[0037] The barrier metal layer 6a' is, for example, a TiN film (titanium nitride film). Each of the metal layers 21 and 22 is, for example, a W layer. According to the present comparative example, the barrier metal layer 6a' and the metal layer 22 are polycrystalline layers and the metal layer 21 is an amorphous layer. Thus, the metal layer 22 is lower in resistivity than the metal layer 21.

[0038] According to the present comparative example, the metal layer 21 is formed using WF.sub.6 gas and B.sub.2H.sub.6 gas and the metal layer 22 is formed using WF.sub.6 gas and H.sub.2 gas (where W is tungsten, F is fluorine, B is boron, and H is hydrogen). Thus, the metal layers 21 and 22 include F atoms as impurity atoms.

[0039] FIG. 6B is a sectional view for explaining a problem with the semiconductor device of the present comparative example. With the present comparative example, if the electrode layers 6 are reduced in thickness to downsize the semiconductor device, it becomes difficult to embed the electrode material layers 6b in the concave portions Hb (see FIG. 5) and easy for large seams Ga to be formed in the electrode material layers 6b. F-atom concentration in the electrode layers 6 is high on surfaces of the electrode layers 6, and thus F atoms in the electrode layers 6 exist at high concentration on the surfaces of the electrode layers 6 facing the seams Ga. This makes it easy for the F atoms in the electrode layers 6 to diffuse through the seams Ga, which might cause failures (e.g., an electrical insulation failure) of the semiconductor device. Thus, it is desired to inhibit diffusion of F atoms by means of the barrier metal layers 6a'. The arrows shown in FIG. 6B indicate how F atoms diffuse from the electrode layer 6 to other layers (e.g., insulators 14).

[0040] FIG. 7 is a sectional view showing the structure of the semiconductor device of the first embodiment.

[0041] As shown in FIG. 7, the electrode layer 6 of the present embodiment includes the barrier metal layer 6a. The electrode material layer 6b of the present embodiment includes metal layers 25 and 26 formed in sequence on a side face, upper face, and lower face of the barrier metal layer 6a. The barrier metal layer 6a is an example of the first layer. The metal layer 25 is an example of the third layer. The metal layer 26 is an example of the second layer.

[0042] The barrier metal layer 6a of the present embodiment includes W (tungsten) atoms and N (nitrogen) atoms. The barrier metal layer 6a is, for example, a W layer including N atoms as impurity atoms as described above. That is, the barrier metal layer 6a includes, for example, W as a principal component. Note that a W layer including N atoms as impurity atoms is denoted as a W(N) layer. As described later, when a composition ratio of W atoms and N atoms in a WN film is 6:4, a ratio k of the number n2 of N atoms to the sum of the number of W atoms n1 and the number n2 of N atoms in the W (N) layer of the present embodiment is lower than 40% (i.e., $k=n2/(n1+n2) < 0.40$). Besides, a ratio of the number of W atoms to the number of all the atoms in the barrier metal layer 6a of the present embodiment is larger than 60%. According to the present embodiment, the barrier metal layer 6a is a crystalline layer, i.e., for example, a polycrystalline layer.

[0043] The metal layer 25 is, for example, a W layer. The metal layer 26 is, for example, a W layer. Each of the metal layers 25 and 26 may or may not include N atoms as impurity atoms. However, N concentration in the barrier metal layer 6a of the present embodiment is higher than N concentration in the metal layer 26 even when the metal layer 26 includes N atoms as impurity atoms. The N concentration in the metal layer 26 is, for example, 7.0×10^{19} to 2.0×10^{20} atoms/cm³. Similarly, N concentration in the barrier metal layer 6a of the present embodiment is higher than N concentration in the metal layer 25 even when the metal layer 25 includes N atoms as impurity atoms. According to the present embodiment, the metal layer 26 is a polycrystalline

layer and the metal layer **25** is an amorphous layer. Thus, the metal layer **26** is lower in resistivity than the metal layer **25**.

[0044] The barrier metal layer **6a** is formed, for example, by forming a WN film by CVD and changing the WN film into a W (N) layer by annealing as described later. According to the present embodiment, the WN film is annealed using a predetermined gas such as H.sub.2 (hydrogen) gas (H.sub.2 annealing). The predetermined gas may be, for example, Ar (argon) gas or N.sub.2 (nitrogen) gas. The annealing is performed, for example, at a temperature higher than 400° C., and desirably at a temperature equal to or higher than 500° C. According to the present embodiment, the WN film, which is an amorphous layer, is crystalized by the annealing and changes into a W(N) layer (the barrier metal layer **6a**), which is a crystalline layer. The annealing is performed, for example, at a temperature higher than 400° C. and lower than 1050° C. Duration of the annealing is, for example, one hour, but another duration may be used. According to the present embodiment, the W(N) layer (barrier metal layer **6a**) with a ratio k of less than 40% is formed from the WN film with a ratio k of 40%, and the N concentration in the W(N) layer becomes lower than the N concentration in the WN film. The WN film is an example of the first layer.

[0045] The WN film used to form the barrier metal layer **6a** is formed using, for example, a source gas including W and a reducing gas including N. The source gas is, for example, WF.sub.6 gas. The reducing gas is, for example, NH.sub.3 (ammonia) gas. In this case, the WN film as well as the W(N) layer (barrier metal layer **6a**) formed of the WN film can include F atoms and H atoms as impurity atoms. The source gas may include a halogen other than F, i.e., for example, Cl (chlorine). For example, when a WN film is formed using WOCl.sub.4 gas and NH.sub.3 gas, the WN film and the W(N) layer can include Cl atoms, O atoms, and H atoms as impurity atoms (where O denotes oxygen). Note that the WN film used to form the barrier metal layer **6a** may be formed by a method other than CVD. For example, the WN film used to form the barrier metal layer **6a** may be formed by ALD (atomic layer deposition).

[0046] The metal layer **25** is formed using, for example, a source gas including W and F, and a reducing gas including B and H. The source gas is, for example, WF.sub.6 gas. The reducing gas is, for example, B.sub.2H.sub.6. In this case, the metal layer **25** may include F atoms, B atoms, and H atoms as impurity atoms. The source gas may include a halogen other than F, i.e., for example, Cl (chlorine). An example of such a source gas is WOCl.sub.4 gas. In this case, the metal layer **25** may include Cl atoms and O atoms as impurity atoms. On the other hand, the reducing gas may include Si instead of B. Examples of such a reducing gas include SiH.sub.4 gas and Si.sub.2H.sub.6 gas. In this case, the metal layer **25** can include Si atoms and H atoms as impurity atoms.

[0047] The metal layer **26** is formed using, for example, a source gas including W and F, and a reducing gas including H. The source gas is, for example, WF.sub.6 gas. The reducing gas is, for example, H.sub.2 gas. In this case, the metal layer **26** can include F atoms and H atoms as impurity atoms. The source gas may include a halogen other than F, i.e., for example, Cl (chlorine). An example of such a source gas is WOCl.sub.4 gas. In this case, the metal layer **26** may include Cl atoms and O atoms as impurity atoms. By forming the electrode layer **6** using the W(N) layer (barrier metal layer **6a**), the present embodiment makes it possible to reduce the metal layer **25** in thickness compared to when the barrier metal layer **6a** is a TiN film. Thus, by reducing the thickness of the metal layer **25**, which is a high-resistance layer, electric resistance of the electrode layer **6** can be reduced. In addition, the W(N) layer is more effective than a WN film in inhibiting the diffusion of F atoms. Thus, by forming the electrode layer **6** using the W (N) layer, the present embodiment makes it possible to effectively inhibit diffusion of F atoms. For example, the F (fluorine) concentration in the insulators **14** in the case of using the W(N) layer becomes equal to or lower than one third of the F concentration in the insulators **14** in the case of using no W(N) layer. It is thereby understood that the W(N) layer suppresses the diffusion of the F atoms. This similarly applies to halogen atoms other than F atoms. Further details on this subject will be

described later.

[0048] FIGS. 8A to 8C are sectional views showing details of the method of manufacturing the semiconductor device of the first embodiment. More specifically, FIGS. 8A to 8C show the process of forming the structure shown in FIG. 7.

[0049] In forming the electrode layer 6 in each concave portion Hb, first, a barrier metal layer 6c is formed on a surface of the insulator 5b (FIG. 8A). The barrier metal layer 6c is, for example, a WN film. The barrier metal layer 6c of the present embodiment is formed by CVD using WF₆ gas and NH₃ gas, and a composition ratio of W atoms and N atoms in the barrier metal layer 6c is 6:4. Consequently, the ratio k of the number n₂ of N atoms to the sum of the number of W atoms n₁ and the number n₂ of N atoms in the barrier metal layer 6c is 40% (i.e., $k=n_2/(n_1+n_2)=0.40$). The barrier metal layer 6c is an example of the first layer.

[0050] Next, the barrier metal layer 6c is annealed (FIG. 8B). The annealing is performed, for example, using a predetermined gas (e.g., H₂ gas, Ar gas or H₂ gas) at a temperature higher than 400° C. The annealing is performed at a temperature of, for example, 500° C. or above, but lower than 1050° C. As a result, the barrier metal layer 6c changes into the barrier metal layer 6a. The barrier metal layer 6a is, for example, a W(N) layer, i.e., a W layer including N atoms as impurity atoms. According to the present embodiment, the WN film (barrier metal layer 6c), which is an amorphous layer, is crystalized by the annealing, and changes into a W(N) layer (the barrier metal layer 6a), which is a crystalline layer. According to the present embodiment, the barrier metal layer 6a with a ratio k of less than 40% is formed from the barrier metal layer 6c with a ratio k of 40%, and the N concentration in the barrier metal layer 6a becomes lower than the N concentration in the barrier metal layer 6c. The barrier metal layer 6a after the annealing may include an N (nitrogen) concentration gradient. In FIG. 8B, the barrier metal layer 6c is annealed in a state that the barrier metal layer 6c is exposed to the concave portion Hb.

[0051] Next, the metal layer 25 is formed on a surface of the barrier metal layer 6a (FIG. 8C). The metal layer 25 is, for example, a W layer including W as a principal component, and is formed using WF₆ gas and B₂H₆ gas. The metal layer 25 of the present embodiment is formed as an amorphous layer at a temperature of, for example, 300° C. or below (preferably 200° C. or below). The metal layer 25 of the present embodiment is formed as an initial film of the electrode material layer 6b. The N concentration in the barrier metal layer 6a of the present embodiment is higher than the N concentration in the metal layer 25.

[0052] Next, the metal layer 26 is formed on a surface of the metal layer 25 (FIG. 8C). The metal layer 26 is, for example, a W layer including W as a principal component, and is formed using WF₆ gas and H₂ gas. The metal layer 26 of the present embodiment is formed, for example, as a polycrystalline layer at a temperature of 450° C. According to the present embodiment, the metal layer 26 is formed at a temperature higher than is the metal layer 25. The metal layer 26 may be formed so as to include the seam Ga. The metal layer 26 is, for example, an LFW (low-fluorine tungsten) layer or a CVD-W (chemical-vapor-deposition tungsten) layer.

[0053] The metal layer 26 of the present embodiment forms the electrode material layer 6b in conjunction with the metal layer 25. The N concentration in the barrier metal layer 6a of the present embodiment is higher than the N concentration in the metal layer 26. The metal layer 26 is formed such that the N concentration will be, for example, 7.0×10^{19} to 2.0×10^{20} atoms/cm³. Note that the N concentration is a value measured, for example, by SIMS (secondary ion mass spectrometry) analysis.

[0054] According to the present embodiment, the metal layer 25 may also be transformed from an amorphous layer to a polycrystalline layer during or after the formation of the metal layer 26. In that case, the metal layer 25 is crystalized by annealing at a temperature of, for example, 600° C. or above, and preferably 750° C. or above, and changes from an amorphous layer to a polycrystalline layer.

[0055] FIGS. 9A and 9B are graphs concerning characteristics of the semiconductor device of the

first embodiment.

[0056] FIG. 9A shows results of XPS (X-ray photoelectron spectroscopy) analysis on the barrier metal layer 6a, where the abscissa represents annealing temperature of the barrier metal layer 6c and the ordinate represents intensity of N atoms (N intensity) in the barrier metal layer 6a. The N intensity decreases with decreases in the N concentration in the barrier metal layer 6a. FIG. 9A shows results of XPS analysis of the W(N) layer when annealing is performed using H.sub.2 gas or Ar (argon) gas as the predetermined gas.

[0057] It can be seen from FIG. 9A that the N concentration in the barrier metal layer 6a becomes low when H.sub.2 gas is used in the annealing and when Ar gas is used in the annealing. Moreover, it can be seen that the N concentration in the barrier metal layer 6a becomes lower when H.sub.2 gas is used in the annealing. Thus, the barrier metal layer 6c of the present embodiment is preferably annealed using H.sub.2 gas. It can also be seen from

[0058] FIG. 9A that when H.sub.2 gas is used to anneal the W(N) layer, the N intensity of the W(N) layer starts to decrease at 400° C. and becomes far lower at 500° C. than at 400° C. Thus, the barrier metal layer 6c of the present embodiment is annealed at a temperature higher than 400° C., and desirably at a temperature of 500° C. or above.

[0059] In FIG. 9A, the barrier metal layer 6a annealed at 500° C. or above is higher in N intensity than an LFW layer (the metal layer 26). In this case, the N concentration in the barrier metal layer 6a is higher than N concentration in the LFW layer. The ratio k of the barrier metal layer 6a of the present embodiment is lower than 40% as described above, and desirably 10% or below. The ratio k of the barrier metal layer 6a of the present embodiment is, for example, 5% or below. As shown in FIG. 9A, annealing at 500° C. or above using H.sub.2 gas can greatly reduce the N concentration in the barrier metal layer 6a, and thus the barrier metal layer 6a with the ratio k equal to or less than 10% or 5% can be realized by annealing at 500° C. or above using H.sub.2 gas.

[0060] FIG. 9B shows results of X-ray scattering of the barrier metal layer 6a, where the abscissa represents scattering angle (2θ) and the ordinate represents scattering intensity. FIG. 9B shows results of X-ray scattering when the barrier metal layer 6c is annealed at 300° C., 400° C., 500° C., 580° C., or 650° C. using H.sub.2 gas. FIG. 9B further shows results of X-ray scattering of the LFW layer (metal layer 26) as a comparative example.

[0061] It can be seen from FIG. 9B that a scattering peak of X-ray scattering is observed when the annealing temperature is 500° C. or above. From this it can be seen that the barrier metal layer 6a of the present embodiment is a crystalline layer, and tungsten included in the barrier metal layer 6a of the present embodiment includes a crystal structure having a (110) plane as a plane orientation. It can be seen from FIG. 9B that the LFW layer (metal layer 26) of the present embodiment is also a crystalline layer.

[0062] FIG. 10 is another graph concerning characteristics of the semiconductor device of the first embodiment.

[0063] FIG. 10 shows results of XPS analysis on the barrier metal layer 6a, where the abscissa represents annealing temperature of the barrier metal layer 6c and the ordinate represents intensity of F atoms (F intensity) in the barrier metal layer 6a. The F intensity decreases with decreases in the F concentration in the barrier metal layer 6a. FIG. 10 shows results of XPS analysis when annealing is performed using H.sub.2 gas or Ar gas as the predetermined gas.

[0064] It can be seen from FIG. 10 that when annealing is performed using H.sub.2 gas, if the annealing is done at 500° C., the F concentration in the barrier metal layer 6a can be decreased to a level almost equal to the F concentration in the LFW layer (metal layer 26). Thus, annealing at 500° C. or above is effective not only in reducing the N concentration in the barrier metal layer 6a, but also in reducing the F concentration in the barrier metal layer 6a.

[0065] FIG. 11 is a sectional view showing a structure of a semiconductor device of a variation of the first embodiment.

[0066] The structure of the semiconductor device (FIG. 11) of the present variation is similar to the

structure of the semiconductor device (FIG. 7) of the first embodiment. However, whereas the electrode material layer **6b** of the first embodiment includes the metal layers **25** and **26**, an electrode material layer **6b** of the present variation includes only the metal layer **26**. The semiconductor device of the present variation can be manufactured, for example, by a method shown in FIGS. **9A** to **9B** by omitting the step of forming the metal layer **25**.

[0067] As described above, each of the electrode layers **6** of the first embodiment includes the barrier metal layer **6a** which is a W(N) layer, and the electrode material layer **6b** (the metal layers **25** and **26**) which is a W layer. Thus, the present embodiment makes it possible to form the electrode layers **6** having suitable characteristics. For example, it becomes possible to reduce the electric resistance of the electrode layers **6** and inhibit diffusion of F atoms from the electrode layers **6** to other layers.

Second Embodiment

[0068] FIGS. **12A** and **12B** are sectional views showing a structure of a semiconductor device of a second embodiment, and a structure of a semiconductor device of a variation of the second embodiment.

[0069] FIG. **12A** shows the semiconductor device of the second embodiment. The semiconductor device of the present embodiment includes an inter layer dielectric **31** formed on the substrate **11** described above, and an interconnect layer **32** formed in the inter layer dielectric **31**. The interconnect layer **32** is, for example, one of a plurality of interconnect layers forming a multilayer interconnect structure of the semiconductor device of the present embodiment and includes one or more interconnects. FIG. **12A** shows one of the interconnects. The interconnect layer **32** may be a plug layer that includes one or more contact plugs or one or more via plugs. The inter layer dielectric **31** is an example of the first insulator. The interconnect layer **32** is an example of the first interconnect layer.

[0070] The interconnect layer **32** shown in FIG. **12A** includes a barrier metal layer **32a** and an interconnect material layer **32b** formed in sequence in the inter layer dielectric **31**. The barrier metal layer **32a** is formed on a side face and upper face of the inter layer dielectric **31**. The interconnect material layer **32b** includes a metal layer **42** formed on a side face and upper face of the barrier metal layer **32a**. FIG. **12A** further shows a seam Gb formed in the metal layer **42** when the metal layer **42** is formed. The seam Gb makes up an air gap extending generally in parallel to the Z direction. The barrier metal layer **32a** is an example of the first layer. The metal layer **42** is an example of the second layer.

[0071] The barrier metal layer **32a** of the present embodiment corresponds to the barrier metal layer **6a** of the first embodiment and is formed by a method similar to the method for the barrier metal layer **6a** of the first embodiment. Thus, the barrier metal layer **32a** is, for example, a W(N) layer, and has an N concentration similar to the N concentration of the barrier metal layer **6a**.

[0072] The metal layer **42** of the present embodiment corresponds to the metal layer **26** of the first embodiment and is formed by a method similar to the method for the metal layer **26** of the first embodiment. Thus, the metal layer **42** is, for example, a W layer, and has an N concentration similar to the N concentration of the metal layer **26**.

[0073] Note that the interconnect layer **32** of the present embodiment does not include a metal layer corresponding to the metal layer **26** of the first embodiment. Thus, the interconnect layer **32** of the present embodiment has a structure similar to the structure of the electrode layer **6** of the variation of the first embodiment shown in FIG. **11**.

[0074] FIG. **12B** shows a semiconductor device of a variation of the second embodiment. A structure of the semiconductor device of the present variation is similar to the structure of the semiconductor device of the second embodiment shown in FIG. **12A**. However, the interconnect material layer **32b** of the present variation includes a metal layer **41** between the barrier metal layer **32a** and the metal layer **42**. The metal layer **41** is an example of the third layer.

[0075] The metal layer **41** of the present variation corresponds to the metal layer **25** of the first

embodiment and is formed by a method similar to the method for the metal layer **25** of the first embodiment. Thus, the metal layer **41** is, for example, a W layer, and has an N concentration similar to the N concentration of the metal layer **25**.

[0076] FIGS. **13A** to **13D** are sectional views showing a method of manufacturing the semiconductor device of the second embodiment.

[0077] First, the inter layer dielectric **31** is formed on the substrate **11**, and an interconnect trench Hc is formed in the inter layer dielectric **31** by photolithography and RIE (FIG. **13A**). The inter layer dielectric **31** may be formed on the substrate **11** either directly or via another layer. The interconnect trench Hc is an example of the first concave portion.

[0078] Next, a barrier metal layer **32c** is formed on a side face and upper face of the inter layer dielectric **31** in the interconnect trench Hc (FIG. **13B**). The barrier metal layer **32c** is, for example, a WN film. The barrier metal layer **32c** of the present embodiment is formed by CVD using WF.sub.6 gas and NH.sub.3 gas, and a composition ratio of W atoms and N atoms in the barrier metal layer **32c** is 6:4. Consequently, the ratio k of the number n_2 of N atoms to the sum of the number of W atoms n_1 and the number n_2 of N atoms in the barrier metal layer **32c** is 40% (i.e., $k = n_2 / (n_1 + n_2) = 0.40$). The barrier metal layer **32c** is an example of the first layer. Note that illustration of the barrier metal layer **32c** formed outside the interconnect trench Hc is omitted.

[0079] Next, the barrier metal layer **32c** is annealed (FIG. **13C**). The annealing is performed, for example, using a predetermined gas (e.g., H.sub.2) at a temperature higher than 400° C. The annealing is performed at a temperature of, for example, 500° C. or above, but lower than 1050° C. As a result, the barrier metal layer **32c** changes into the barrier metal layer **32a**. The barrier metal layer **32a** is, for example, a W(N) layer, i.e., a W layer including N atoms as impurity atoms. According to the present embodiment, the WN film (barrier metal layer **32c**), which is an amorphous layer, is crystalized by the annealing, and changes into a W(N) layer (the barrier metal layer **32a**), which is a crystalline layer. According to the present embodiment, the barrier metal layer **32a** with a ratio k of less than 40% is formed from the barrier metal layer **32c** with a ratio k of 40%, and the N concentration in the barrier metal layer **32a** becomes lower than the N concentration in the barrier metal layer **32c**.

[0080] Next, the metal layer **42** (interconnect material layer **32b**) is formed on a side face and upper face of the barrier metal layer **32a** in the interconnect trench Hc (FIG. **13C**). The metal layer **42** is, for example, a W layer including W as a principal component, and is formed using WF.sub.6 gas and H.sub.2 gas. The metal layer **42** of the present embodiment is formed, for example, as a polycrystalline layer at a temperature of 450 to 2000 C. The metal layer **42** may be formed so as to include the seam Gb. The metal layer **42** is, for example, an LFW layer or a CVD-W layer. The N concentration in the barrier metal layer **32a** of the present embodiment is higher than the N concentration in the metal layer **42**. The metal layer **42** is formed such that the N concentration will be, for example, 7.0×10^{19} to 2.0×10^{20} atoms/cm³. Note that illustration of the metal layer **42** formed outside the interconnect trench Hc is omitted.

[0081] Subsequently, the barrier metal layer **32a** and metal layer **42** formed outside the interconnect trench Hc are removed, for example, by CMP (chemical mechanical polishing). As a result, the interconnect layer **32** including the barrier metal layer **32a** and metal layer **42** is formed.

[0082] In this way, the semiconductor device of the second embodiment shown in FIG. **12A** is manufactured. Note that in manufacturing the semiconductor device of the variation of the second embodiment shown in FIG. **12B**, the metal layer **41** is formed before the formation of the metal layer **42**, but after the formation of the barrier metal layer **32a**. The method of forming the metal layer **41** is similar to the method of forming the metal layer **25** in the step of FIG. **8C**.

[0083] As described above, the interconnect layer **32** of the present embodiment includes the barrier metal layer **32a**, which is a W(N) layer, and the interconnect material layer **32b**, which is a W layer. Thus, the present embodiment makes it possible to form the interconnect layers **32** having suitable characteristics. For example, as with the electrode layers **6** of the first embodiment, it

becomes possible to reduce the electric resistance of the interconnect layers **32** and inhibit diffusion of F atoms from the interconnect layers **32** to other layers.

Third Embodiment

[0084] FIG. **14** is a sectional view showing a structure of a semiconductor device of a third embodiment. The semiconductor device of the present embodiment is, for example, a three-dimensional semiconductor memory, and is manufactured by laminating an array wafer including an array region **101** with a circuit wafer including a circuit region **102**. Relationships between the present embodiment and the first and second embodiments will be described later.

[0085] The array region **101** includes a memory cell array **111** including a plurality of memory cells, an insulator **112** above the memory cell array **111**, and an inter layer dielectric **113** under the memory cell array **111**. The insulator **112** is, for example, a silicon oxide film or a silicon nitride film. The inter layer dielectric **113** is, for example, a silicon oxide film, or a stacked film including a silicon oxide film and another insulator.

[0086] The circuit region **102** is provided under the array region **101**. FIG. **14** shows a boundary face (bonding face) S between the array region **101** and the circuit region **102**. The circuit region **102** includes an inter layer dielectric **114** under the inter layer dielectric **113** and a substrate **115** under the inter layer dielectric **114**. The inter layer dielectric **114** is, for example, a silicon oxide film, or a stacked film including a silicon oxide film and another insulator. The substrate **115** is a semiconductor substrate such as a silicon substrate.

[0087] FIG. **14** shows an X direction and a Y direction parallel to a surface of the substrate **115** and perpendicular to each other, and a Z direction perpendicular to the surface of the substrate **115**. In the present specification, a +Z direction is treated as an upward direction and a -Z direction is treated as a downward direction. The -Z direction may or may not coincide with the gravity direction.

[0088] The array region **101** includes a plurality of word lines WL and a source line SL as a plurality of electrode layers in the memory cell array **111**. FIG. **14** shows a staircase-like structure **121** of the memory cell array **111**. Each of the word lines WL is electrically connected with a word line layer **123** via a contact plug **122**. Each of columnar portions CL penetrating the plurality of word lines WL is electrically connected with a bit line BL via a via plug **124** and electrically connected with the source line SL. The source line SL includes a lower layer SL1, which is a semiconductor layer and an upper layer SL2, which is a metal layer.

[0089] The circuit region **102** includes a plurality of transistors **131**. Each of the transistors **131** includes a gate electrode **132** provided on the substrate **115** via a gate insulator as well as a non-illustrated source diffusion layer and drain diffusion layer provided in the substrate **115**. The circuit region **102** further includes the gate electrodes **132** of the transistors **131**, a plurality of contact plugs **133** provided on the source diffusion layer or the drain diffusion layer, and interconnect layers **134** provided on the contact plugs **133** and including a plurality of interconnects.

[0090] The circuit region **102** includes an interconnect layer **135** provided above the interconnect layers **134** and including a plurality of interconnects; and an interconnect layer **136** provided above the interconnect layer **135** and including a plurality of interconnects. The circuit region **102** further includes a plurality of via plugs **137** provided on the interconnect layer **136** and a plurality of metal pads **138** provided on the via plugs **137**. The metal pads **138** are, for example, metal layers including Cu (copper) layers. The circuit region **102** functions as a control circuit (logic circuit) adapted to control operation of the array region **101**. The control circuit is made up of the transistors **131** and the like and electrically connected to the metal pads **138**.

[0091] The array region **101** includes a plurality of metal pads **141** provided on the metal pads **138**, and a plurality of via plugs **142** provided on the metal pads **141**. The metal pads **141** are, for example, metal layers including Cu layers. The array region **101** further includes interconnect layers **143** provided on the via plugs **142** and including a plurality of interconnects; and an interconnect layer **144** provided on the interconnect layers **143** and including a plurality of

interconnects. The bit line BL is included in the interconnect layer **144**. The control circuit is electrically connected to the memory cell array **111** via the metal pads **141**, **138**, and the like, and controls operation of the memory cell array **111** via the metal pads **141**, **138**, and the like.

[0092] The array region **101** also includes a plurality of via plugs **145** provided on the interconnect layer **144**, a metal pad **146** provided on the via plugs **145** and the insulator **112**, and a passivation insulator **147** provided on the metal pad **146** and the insulator **112**. The metal pad **146** is, for example, a metal layer including a Cu layer and functions as an external connection pad (bonding pad) of the semiconductor device of the present embodiment. The passivation insulator **147** is, for example, a stacked film including a silicon oxide film and a silicon nitride film and has an opening P that exposes an upper face of the metal pad **146**. Through the opening P, the metal pad **146** can be electrically connected to a mounted substrate or another device via a bonding wire, a solder ball, a metal bump, or the like.

[0093] FIG. **15** is a sectional view showing a structure of a columnar portion CL of the third embodiment. FIG. **15** shows one of a plurality of columnar portions shown in FIG. **14**.

[0094] As shown in FIG. **15**, the memory cell array **111** includes the plurality of word lines WL and a plurality of insulators **151** stacked alternately on the inter layer dielectric **113** (see FIG. **14**). The word lines WL are, for example, metal layers including W layers or the like. The insulators **151** are, for example, silicon oxide films.

[0095] The columnar portion CL includes a block insulator **152**, a charge storage layer **153**, a tunnel insulator **154**, a channel semiconductor layer **155**, and a core insulator **156** in sequence. The charge storage layer **153** is, for example, an insulator such as a silicon nitride film, and is formed on side faces of the word lines WL and insulators **151** via the block insulator **152**. The channel semiconductor layer **155** is, for example, a polysilicon layer, and is formed on a side face of the charge storage layer **153** via the tunnel insulator **154**. The block insulator **152**, the tunnel insulator **154**, and the core insulator **156** are, for example, silicon oxide films.

[0096] FIGS. **16** and **17** are sectional views showing a method of manufacturing the semiconductor device of the third embodiment.

[0097] FIG. **16** shows an array wafer W1 and a circuit wafer W2. Specifically, FIG. **16** shows one of a plurality of the array regions **101** included in the array wafer W1 and one of a plurality of the circuit regions **102** included in the circuit wafer W2.

[0098] The array wafer W1 in FIG. **16** is opposite in direction to the array region **101** in FIG. **14**. According to the present embodiment, a plurality of the semiconductor devices are manufactured by bonding together the array wafer W1 and the circuit wafer W2. FIG. **16** shows the array wafer W1 before being reversed in direction for bonding and FIG. **14** shows the array region **101** after being bonded and diced by being reversed in direction for bonding.

[0099] FIG. **16** further shows an upper face S1 of the array wafer W1 and an upper face S2 of the circuit wafer W2. The array wafer W1 includes a substrate **116** provided under the insulator **112**. The substrate **116** is a semiconductor substrate such as a silicon substrate.

[0100] According to the present embodiment, first, as shown in FIG. **16**, the memory cell array **111**, the insulator **112**, the inter layer dielectric **113**, the staircase-like structure **121**, the metal pads **141**, and the like are formed on the substrate **116** of the array wafer W1, and the inter layer dielectric **114**, the transistors **131**, the metal pads **138**, and the like are formed on the substrate **115** of the circuit wafer W2. For example, the via plugs **145**, the interconnect layer **144**, the interconnect layers **143**, the via plugs **142**, and the metal pads **141** are formed in sequence on the substrate **116**. The contact plugs **133**, the interconnect layers **134**, the interconnect layer **135**, the interconnect layer **136**, the via plugs **137**, and the metal pads **138** are formed on the substrate **115**. Next, as shown in FIG. **17**, the array wafer W1 and the circuit wafer W2 are bonded together by mechanical pressure. Consequently, the inter layer dielectric **113** and the inter layer dielectric **114** are bonded together. Next, the array wafer W1 and the circuit wafer W2 are annealed. Consequently, the metal pads **141** and the metal pads **138** are joined together.

[0101] Subsequently, the substrate **115** is reduced in thickness by CMP and the substrate **116** is removed by CMP, and then the array wafer **W1** and the circuit wafer **W2** are diced into a plurality of chips. In this way, the semiconductor device shown in FIG. **14** is manufactured. Note that the metal pad **146** and the passivation insulator **147** are formed on the insulator **112**, for example, after the substrate **115** is reduced in thickness and the substrate **116** is removed.

[0102] Note that whereas FIG. **14** shows a boundary face between the inter layer dielectric **113** and the inter layer dielectric **114** as well as a boundary face between the metal pads **141** and the metal pads **138**, it is generally the case that the boundary faces can no longer be observable after the annealing. However, locations where the boundary faces existed can be estimated by detecting, for example, inclinations of side faces of the metal pads **141** and inclinations of side faces of the metal pads **138**, and displacement between the side faces of the metal pads **141** and the side faces of the metal pads **138**.

[0103] Next, relationships between the present embodiment and the first and second embodiments will be described.

[0104] The word lines WL of the present embodiment may be formed, for example, by a method similar to the method for the electrode layers **6** of the first embodiment. This makes it possible to reduce the electric resistance of the word lines WL and inhibit diffusion of F atoms from the word lines WL to other layers. In this case, the substrate **116**, insulators **151**, block insulator **152**, charge storage layer **153**, tunnel insulator **154**, channel semiconductor layer **155**, and core insulator **156** of the present embodiment correspond respectively to the substrate **11**, insulator **14**, block insulator **5**, charge storage layer **4**, tunnel insulator **3**, channel semiconductor layer **2**, and core insulator **1** of the first embodiment. However, the substrate **116** of the present embodiment is removed after the array wafer **W1** and the circuit wafer **W2** are bonded together, does not remain in the manufactured semiconductor device.

[0105] At least any of the interconnect layers **134**, **135**, **136**, **143**, and **144** of the present embodiment may be formed, for example, by a method similar to the method for the interconnect layer **32** of the second embodiment. This will make it possible to reduce electric resistance of the interconnect layer(s) and inhibit diffusion of F atoms from the interconnect layer(s) to other layers. In this case, the substrate **116** or substrate **115** of the present embodiment corresponds to the substrate **11** of the second embodiment while the inter layer dielectric **114** or inter layer dielectric **113** corresponds to the inter layer dielectric **31** of the second embodiment. However, the substrate **116** of the present embodiment is removed after the array wafer **W1** and the circuit wafer **W2** are bonded together, and thus does not remain in the manufactured semiconductor device.

[0106] Thus, as with the first and second embodiments, the present embodiment makes it possible to form interconnect layers having suitable characteristics.

[0107] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel devices and methods described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the devices and methods described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A method of manufacturing a semiconductor device, comprising: forming a first insulator; forming a first layer including tungsten and nitrogen, on the first insulator; reducing nitrogen concentration in the first layer by annealing the first layer in a state that the first layer is exposed; and forming a second layer including tungsten, on the first layer after the annealing.
2. The method of claim 1, wherein the annealing is performed at a temperature higher than 400° C.

and lower than 1050° C.

3. The method of claim 1, wherein the annealing is performed using H.sub.2 (hydrogen) gas.

4. The method of claim 1, wherein the annealing is performed using Ar (argon) gas or N.sub.2 (nitrogen) gas.

5. The method of claim 1, wherein the first layer before the annealing is a WN film (tungsten nitride film).

6. The method of claim 1, wherein the first layer is formed by CVD (chemical vapor deposition).

7. The method of claim 1, wherein the first layer is crystalized by the annealing.

8. The method of claim 1, wherein tungsten included in the first layer after the annealing includes a crystal structure having a (110) plane.

9. The method of claim 1, wherein the first layer and the second layer are formed such that nitrogen concentration in the first layer after the annealing is higher than nitrogen concentration in the second layer, and that a ratio of a number of nitrogen atoms to a sum of a number of tungsten atoms and the number of nitrogen atoms in the first layer after the annealing is lower than 40%.

10. The method of claim 1, further comprising: forming a stacked film alternately including a plurality of fourth layers and a plurality of second insulators; forming a charge storage layer on side faces of the plurality of fourth layers via a third insulator; and forming a semiconductor layer on a side face of the charge storage layer via a fourth insulator, removing the plurality of fourth layers to form a plurality of first concave portions in the stacked film; and forming a plurality of electrode layers in the plurality of first concave portions, wherein the first layer and the second layer are one electrode layer of the plurality of electrode layers.

11. A semiconductor device comprising: a first insulator; and a first interconnect layer including a first layer that is provided on the first insulator and includes tungsten and nitrogen, and a second layer wherein nitrogen concentration in the first layer is higher than nitrogen tungsten included in the first layer includes a crystal structure having a (110) plane.

12. The device of claim 11, wherein the first layer includes tungsten as a principal component, and includes nitrogen atoms as impurity atoms.

13. The device of claim 11, wherein a ratio of a number of nitrogen atoms to a sum of a number of tungsten atoms and the number of nitrogen atoms in the first layer is lower than 40%.

14. The device of claim 11, wherein the nitrogen concentration in the second layer is 7.0×10^{19} to 2.0×10^{20} atoms/cm³.

15. The device of claim 11, wherein the second layer further includes fluorine or chlorine.

16. The device of claim 11, wherein the first layer is a barrier metal layer, and the second layer is an interconnect material layer.

17. The device of claim 11, wherein the first interconnect layer further includes a third layer that is provided between the first layer and the second layer and includes tungsten.

18. The device of claim 17, wherein the third layer further includes boron or silicon.

19. A semiconductor device comprising: a stacked film alternately including a plurality of electrode layers and a plurality of second insulators; a charge storage layer provided on side faces of the plurality of electrode layers via a third insulator; and a semiconductor layer provided on a side face of the charge storage layer via a fourth insulator. wherein an electrode layer of the plurality of electrode layers includes a first layer that includes tungsten and nitrogen, and a second layer that is provided on the first layer and includes tungsten, nitrogen concentration in the first layer is higher than nitrogen concentration in the second layer, and tungsten included in the first layer includes a crystal structure having a (110) plane.

20. The device of claim 19, wherein the electrode layer of the plurality of electrode layers further includes a third layer that is provided between the first layer and the second layer and includes tungsten.
